

Stat-X498/S690: Statistical Consulting

Meeting time: Tu 3:55-6:25pm

Meeting location: SW103

Website: <http://jfukuyama.github.io/teaching/stat498>

Instructor: Prof. Julia Fukuyama

jfukuyam at iu dot edu

Office hours: Mondays and Fridays, 12:45-1:45pm

1 Course description

The objectives of this course are to allow students to practice the following skills:

- Translating client requests into statistical analysis plans.
- Implementing analysis plans.
- Communicating results verbally, in writing, and graphically.

2 Class format and weekly structure

In the class sessions, we will work through a series of mock consulting projects. Each week will have roughly the following format.

2.1 Simulated client meeting

Each project begins with a simulated client meeting. The instructor will play the role of the client, and one student will act as the consultant. The goal of this meeting is to clarify the client's question, the available data, and any practical or scientific constraints.

By the end of this segment, the class should share a working understanding of:

- The client's goal
- The quantities or decisions of interest
- The nature of the data-generating process as understood by the client

2.2 Analysis idea generation

Following the client meeting, we will do think-pair-share to generate possible analysis strategies. At this stage, the focus is on breadth: identifying multiple defensible approaches rather than selecting a single plan. We will then think through the pros and cons of each and come up with an implementation plan.

2.3 In-class implementation

Once a common conceptual framework has been established, the class will implement one or more analysis strategies collaboratively. This work may include exploratory analysis, data cleaning, modeling, and diagnostics.

2.4 Interpretation and reflection

At the end of the class session, students will have time to write and submit a brief reflection or code contribution via canvas.

3 Final Project

Each student will select one of the class projects as the basis for a final presentation and written report. Final presentations will take place during the final week of class, and written reports will be due on the date scheduled for the final exam.

4 Course schedule

- Week 1: Introductions, syllabus. Testing out Jupyter Real-Time collaboration.
- Weeks 2-14: Mock projects.
- Weeks 15: Final presentations.

5 Assessment

Grades will be based on:

- Participation: 50%. Each student is expected to contribute analysis ideas, either conceptual (what sorts of analyses should be done in principle) or concrete (what code should we run).
- Final presentation: 15%, scheduled for the last week of class.
- Final project writeup: 35%, due Thursday, May 7 (the day that is scheduled for our final exam). There will be no final exam, only the writeup for the project. You are not prohibited from using generative AI, but points will be taken off for every incorrect statement present in the report. Be sure to only write things that you really believe are true.

6 Policy on incompletes

A grade of Incomplete (I) may be assigned only when a student has completed the majority of the coursework with a passing grade and is unable to complete the remaining portion of the course due to circumstances beyond their control (such as serious illness or family emergency).

An Incomplete cannot be used to avoid a low or failing grade or to allow extra time for additional work beyond the regular course requirements. To request an Incomplete, the student must:

- Contact the instructor as soon as possible, and before the end of the term, to explain the situation and discuss eligibility.
- If the instructor agrees, both the student and instructor will document the remaining work to be completed and the deadline for completion.

If the required work is not completed by the agreed-upon deadline, the Incomplete will automatically lapse to the grade the student earned based on work submitted (often an F if major work is missing).

All Incompletes are subject to the IU policy on incomplete grades.